HR Employee Attrition Analysis

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- Analyzed an HR dataset of 1,481 employee records across 38 attributes to identify key drivers of employee attrition.
- Analyze attrition by age group, gender, education field, job role, salary slab, and satisfaction metrics.
- Calculated attrition rate and key KPIs using DAX, enabling gender-wise and category-wise comparison of attrition trends.
- Derived actionable insights and retention recommendations by correlating attrition with salary levels, job satisfaction, and work environment factors.

OVER ALL KPIs

- The organization has a total workforce of **1,480 employees**, out of which **889 are male employees** and **591 are female employees**, indicating a **male-dominant workforce composition**.

- Out of the total workforce, **238 employees** have exited the organization, comprising **151 male employees** and **87 female employees**, resulting in an overall **attrition rate of 16.08%.**

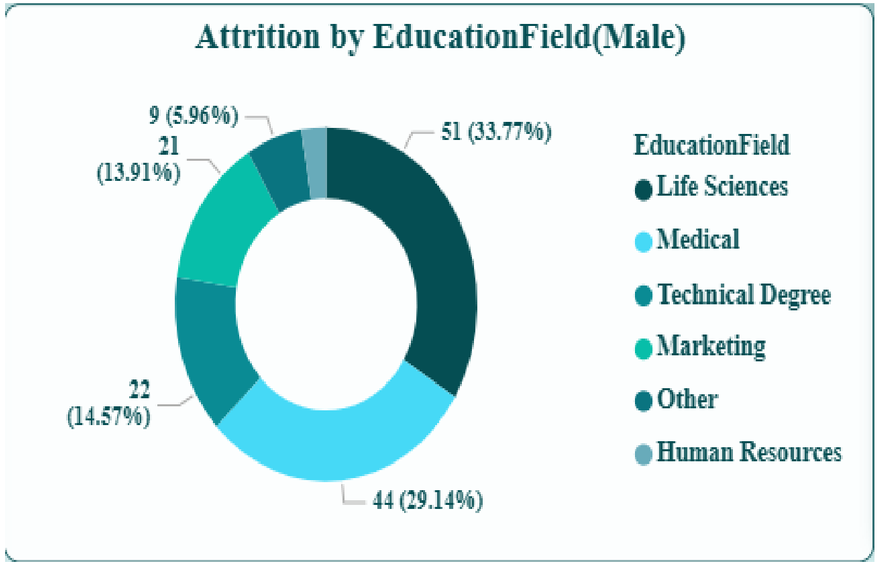
- The **male attrition** rate stands at **16.99%**, which is **higher compared to the female attrition rate of 14.72%**, suggesting that male employees are leaving the organization at a slightly higher rate.

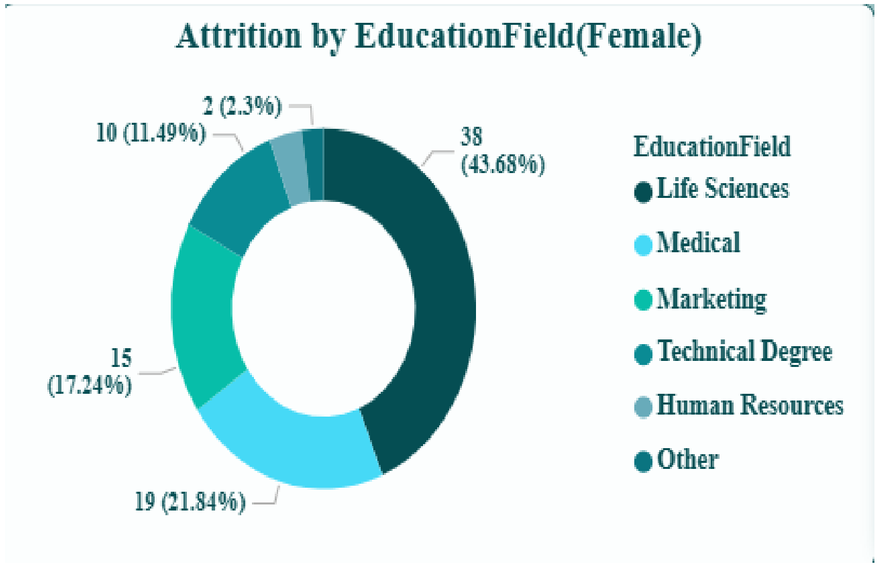
- The **average employee age** across the organization is approximately **37 years**, reflecting a **workforce largely made up of mid-career professionals**.

- The overall **average monthly salary is around 6.5K**, with **female employees** earning a slightly higher average salary**(6.68K)** compared to **male employees(6.39K)**.

- Employees have an **average tenure of approximately 7 years**, indicating a relatively experienced workforce with long-term organizational exposure..

Attrition by Education Field





- Employees from the **Life Sciences** background contribute the highest attrition, with **89 total exits, comprising 51 male and 38 female employees**, **indicating higher mobility** among professionals from this field.

- The **Medical education field** records the second-highest attrition, **totaling 63 employees,** including **44 male and 19 female employees**, suggesting **notable turnover in healthcare-related roles**.

- Attrition among employees with a **Marketing Degree** stands at **36**, with **21 male and 15 female exits**, reflecting **moderate attrition in Marketing roles**.

- The **Technical Degree** accounts for **32 total attritions,** consisting of **22 male and 10 female employees**, indicating comparatively lower but consistent turnover.

- Employees categorized under **Other education fields show 11 attritions**, **where 9 are male and 2 are female**, representing minimal impact on overall attrition.

- The **Human Resources** education field experiences the **lowest attrition**, with only **7 total exits**, including **4 male and 3 female employee**s, highlighting **strong retention within HR-related roles**.

Insight: Attrition is predominantly concentrated among employees from Life Sciences and Medical backgrounds, while Human Resources shows the strongest retention across both male and female employees.

Attrition by Age Group

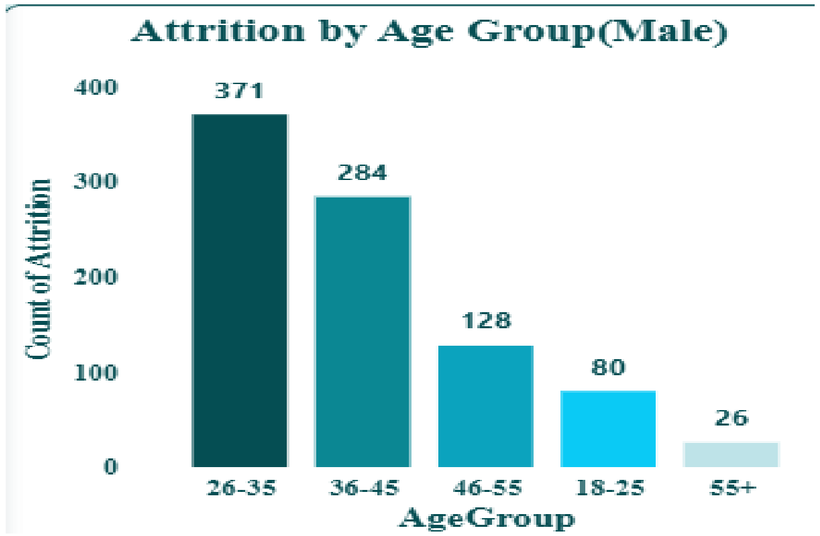
- The **26–35 age group records the highest attrition**, with a total of **611 employees** leaving the organization, out of which **371 are male employees** and **240 are female employees**. This indicates that **early to mid-career professionals form the most vulnerable segment** in terms of retention.

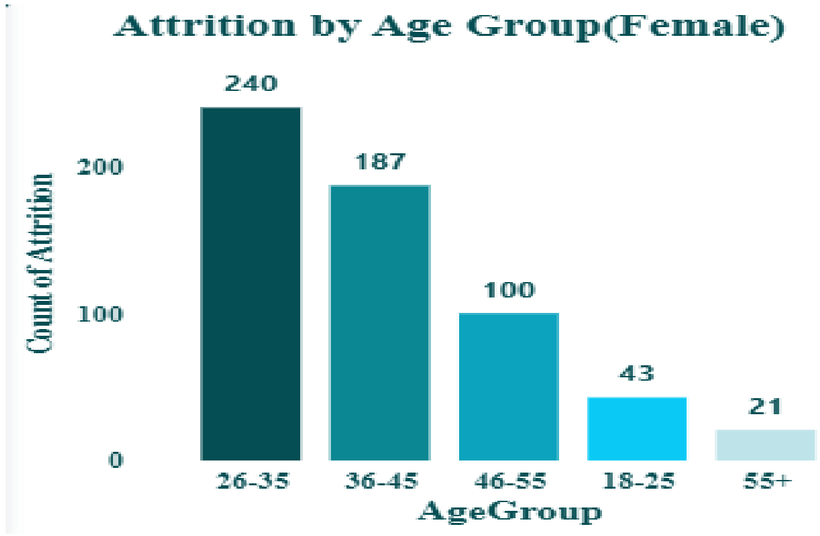
- The **36–45 age group shows the second-highest attrition**, totaling **471 exits**, comprising **284 male and 187 female employees**, suggesting continued attrition pressure among experienced professionals.

- **Attrition declines significantly in the 46–55 age group**, with **228 total exits**, including **128 male and 100 female employees**, indicating greater job stability as **employees progress toward senior roles**.

- The **18–25 age group accounts for 123 attritions**, where **80 are male and 43 are female**, reflecting relatively **lower turnover among entry-level employees**.

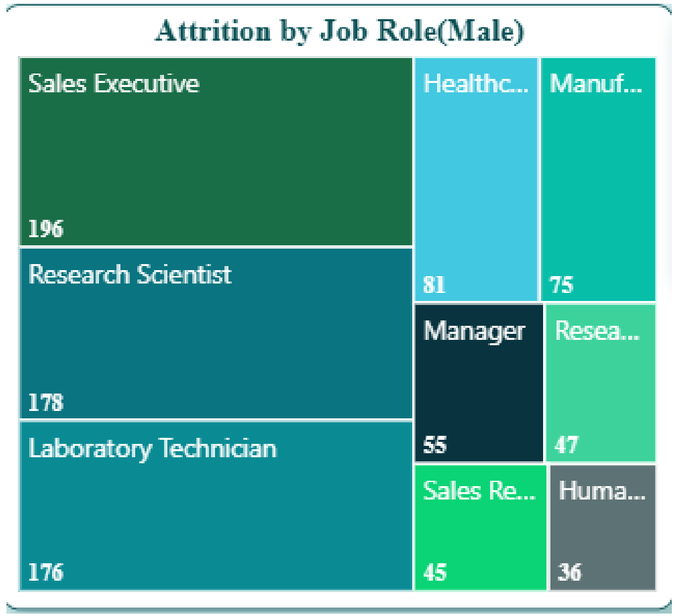
- The **55+ age group experiences the lowest attrition**, with only **47 total exits (26 male and 21 female)**, highlighting strong retention among senior and late-career employees.

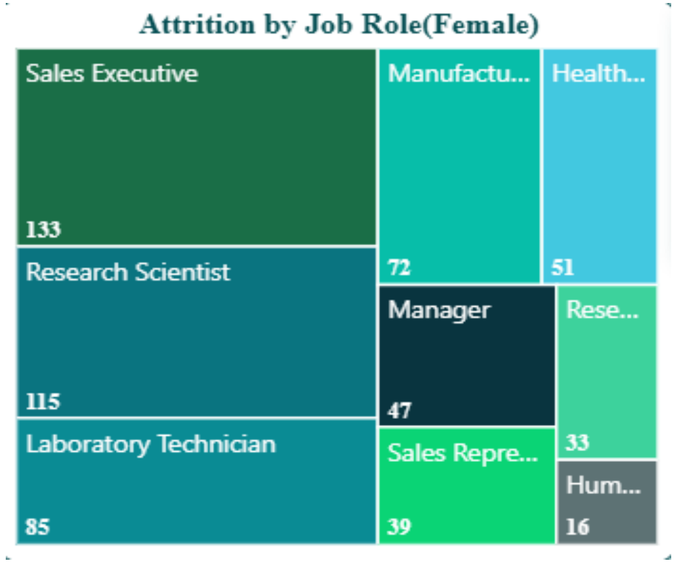




Insights: Attrition is heavily concentrated among mid-career employees aged 26–45, particularly males, indicating a need for targeted retention strategies focused on career growth, compensation, and work–life balance for this segment.

Attrition by Job Role





- **Sales Executives record the highest attrition**, with 329 total exits, comprising 196 male and 133 female employees, indicating high turnover in revenue-driven and target-oriented roles.
- Research Scientists account for the second-highest attrition, totaling 293 employees, including 178 males and 115 females, suggesting retention challenges in research-focused positions.
- Attrition among Laboratory Technicians stands at 261, with 176 male and 85 female exits, reflecting notable turnover in operational and technical support roles.
- The role of Manufacturing Director shows 147 attritions, comprising 75 male and 72 female employees, indicating a relatively balanced gender distribution in exits.
- Managers contribute 102 attritions, where 55 are male and 47 are female, suggesting comparatively better stability at managerial levels.
- Research Directors experience 80 attritions, with 47 male and 33 female employees, indicating lower turnover in senior research leadership roles.
- The Healthcare Representative role records 132 attritions, including 81 male and 51 female employees, highlighting moderate turnover in client-facing healthcare roles.
- Sales Representatives account for 84 attritions, comprising 45 males and 39 females, reflecting consistent turnover in entry-to-mid-level sales roles.
- Human Resources shows the lowest attrition, with only 52 exits, including 36 male and 16 female employees, indicating strong retention within HR functions.

Insight: Attrition is heavily concentrated in sales, research, and technical roles, while managerial and human resources positions demonstrate comparatively stronger workforce stability.

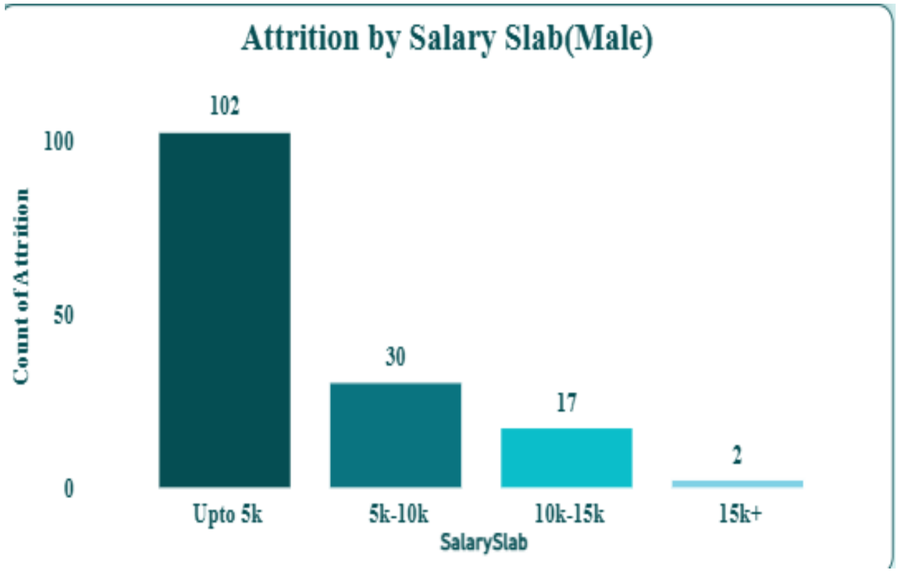
Attrition by Salary Slab

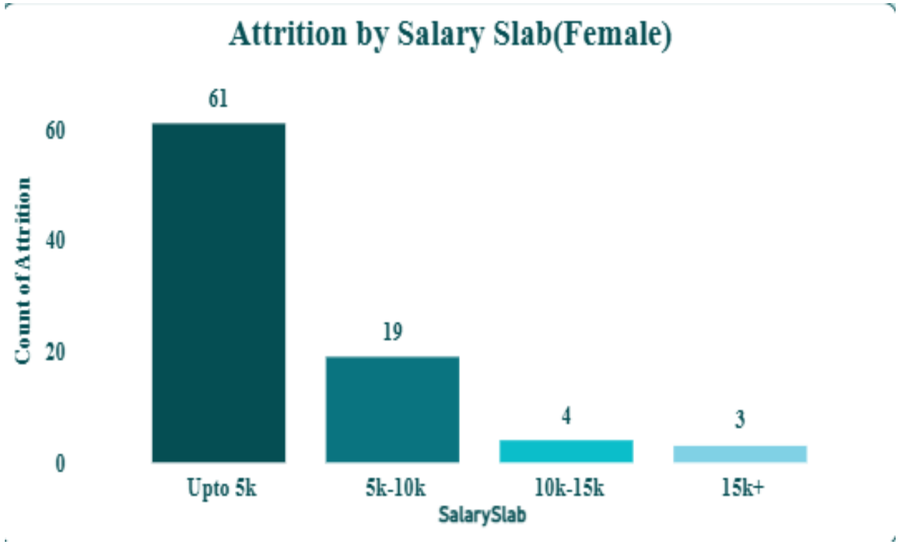
- The "**Up to 5K**" salary slab records the **highest attrition**, with **163 total exits**, comprising **102 male and 61 female employees**, indicating that **lower compensation levels are a major contributor to employee turnover**.

- Employees earning **5K–10K account** for **49 attritions**, including **30 male and 19 female exits**, showing a **significant reduction in attrition as salary levels increase**.

- Attrition further **declines in the 10K–15K salary slab**, with only **21 total exits**, consisting of **17 male and 4 female employees**, reflecting **improved retention among higher-paid employees**.

- The **15K+ salary slab experiences the lowest attrition**, with just **5 total exits, including 2 male and 3 female employees**, highlighting **strong workforce stability at senior compensation levels**.





Employee attrition is heavily concentrated in lower salary slabs and decreases sharply as compensation increases, underscoring salary as a key driver of workforce retention.

Environment Satisfaction

- The **environment satisfaction** analysis indicates a **total score of 4,032**, reflecting the cumulative impact of workplace environment on employee attrition across the organization.

- Out of the total environment satisfaction impact, **male employees contribute a score of 2,422, while female employees contribute 1,610**, indicating a **higher environment-related attrition influence among male employees**.

Job Satisfaction

- The **job satisfaction analysis** shows a **total score of 4,033**, indicating the overall impact of job satisfaction levels on employee attrition across the organization.

- Out of this total, **male employees contribute a score of 2,447**, while **female employees contribute 1,586**, highlighting a comparatively **higher job-satisfaction-related attrition influence among male employees**.

Recommendations

Short-Term Recommendations

- Conduct stay **interviews for high-attrition age groups (26–45),** especially **males**.
- Provide **immediate pay corrections or incentives** for employees in the **lowest salary slab (≤5K)**.
- **Increase manager check-ins and support in high-attrition roles** like **Sales and Research**.
- **Improve quick environment factors** such as team culture, workload balance, and manager support.

Mid-Term Recommendations

- **Define clear career paths** and upskilling plans for mid-career employees.
- **Align compensation bands with market benchmarks** to reduce pay-driven exits.
- **Improve job satisfaction through better role clarity and performance feedback**.
- **Launch gender-focused engagement** initiatives to address higher male attrition.

Long-Term Recommendations

- **Build predictive attrition models** using salary, role, age, and satisfaction data.
- **Invest in leadership** and **people-manager development programs**.
- Create a structured **long-term compensation growth framework**.
- **Institutionalize employee experience** and **engagement programs**.

HR Employee Attrition Analysis

Female | Male

1480
Total Employees

238
Total Attrition

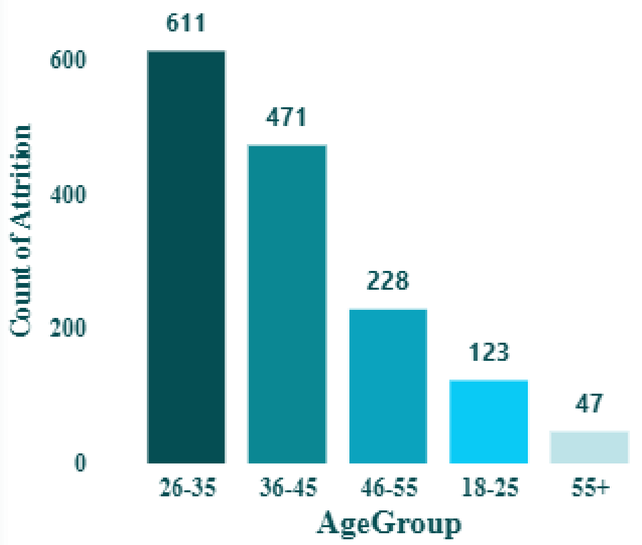
16.08%
Attrition Rate

36.92
Average Age

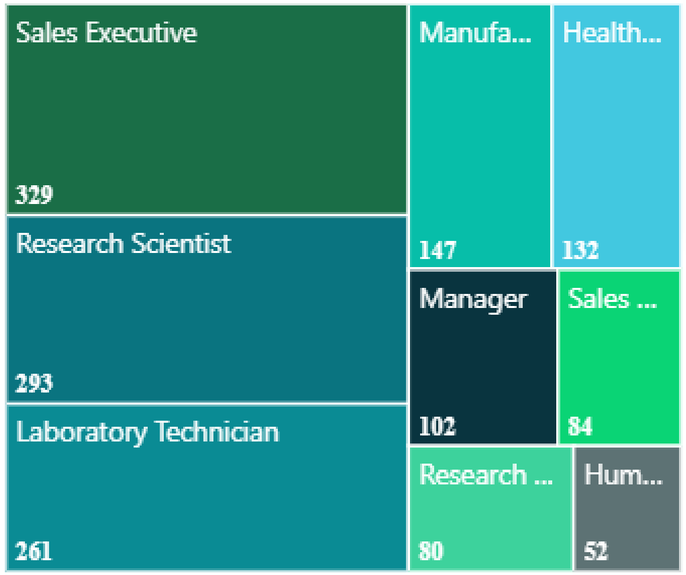
6.50K
Average Salary

7.01
Average Tenure

Attrition by Age Group(Female)



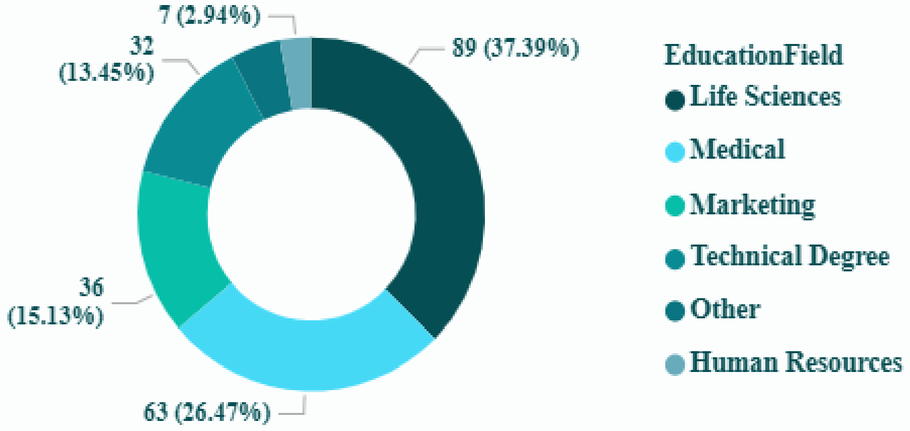
Attrition by Job Role(Male)



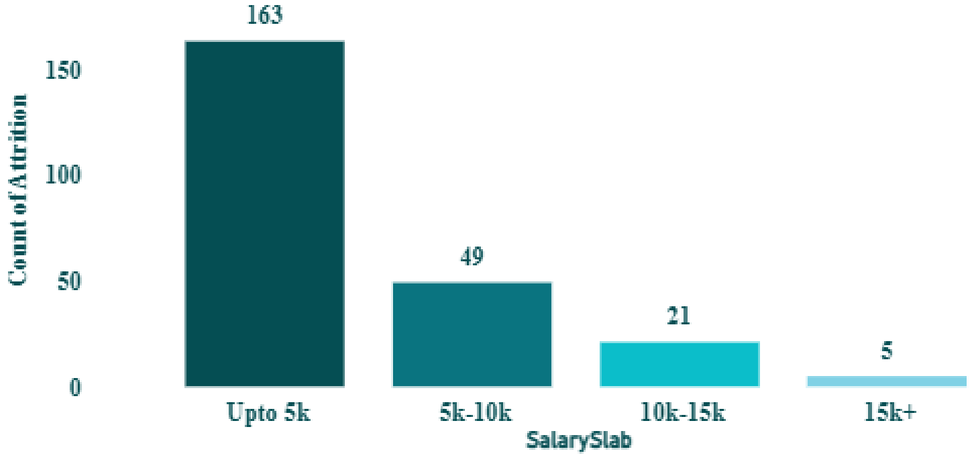
Environment Satisfaction

Job Satisfaction(Female)

Attrition by EducationField(Male)



Attrition by Salary Slab



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